Spontaneous symmetry breaking in two-channel asymmetric exclusion processes with narrow entrances

Reproduction of Pronina & Kolomeisky, J. Phys. A 40, 2275 (2007)

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Outline

Why study exclusion processes?

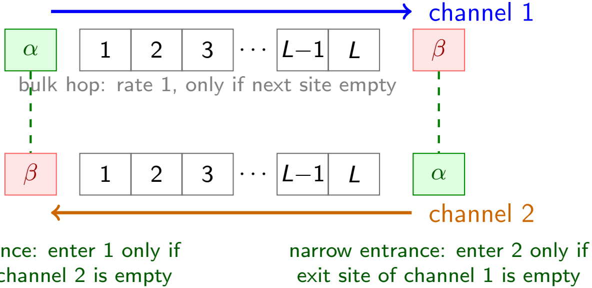
- Asymmetric Simple Exclusion Process (TASEP) is the reference model of *non-equilibrium* transport — the role of the Ising model for equilibrium statistical mechanics.
- Minimal description of interacting particles on a 1D lattice with hard-core exclusion.
- Applications: motor proteins on cytoskeletal filaments (kinesin/dynein), ribosome motion in mRNA translation, car traffic, nanopore transport.

Motivation for two channels

In cells, kinesin and dynein walk *in opposite directions* on the same microtubule. Two lanes of particles moving in opposite directions, coupled at the boundaries — can they *break symmetry* spontaneously?.

The paper's question

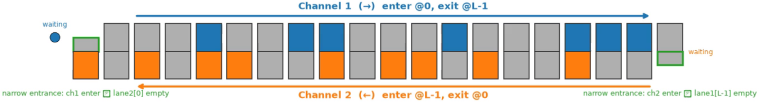
Two identical lanes, identical dynamics, identical injection.



Surprising result

Despite exact symmetry of the rules, the steady state can be **asymmetric**: one lane dense, the other dilute.

The model: two-channel ASEP with narrow entrances



1 frame = 1 event. Bulk hop: rate 1, one site, only if next site empty. Ends: enter α=0.90, exit β=0.30. A waiting dot = entrance blocked by the other lane's exit.

- Channel 1: particles enter at the left, hop right (rate 1), exit right.
- Channel 2: particles enter at the right, hop left (rate 1), exit left.
- Hard-core exclusion: at most one particle per site.

The narrow-entrance rule

- Particle enters channel 1 (rate α) iff site 1 of channel 1 is *empty* **and** the *exit site of channel 2* is empty.
- Particle enters channel 2 (rate α) iff site L of channel 2 is *empty* **and** the *exit site of channel 1* is empty.
- Bulk hops: rate 1 to the next site, only if empty.
- Exits: rate β, *independent* of the other channel.

The coupling

Each channel's *entrance* is coupled to the other channel's *exit*. This is the “narrow entrance” (a bottleneck shared by both lanes at each end).

What we want to compute

Steady-state properties as a function of (α, β):

- Phase diagram in the (α, β) plane.
- Bulk densities ρ_1, ρ_2 and currents J_1, J_2.
- The joint density distribution $P(\rho_1, \rho_2)$ — the diagnostic for spontaneous symmetry breaking (SSB).

Approach: mean-field theory (solves the model analytically), then Monte Carlo (checks the theory).

Recap: single-lane TASEP

One channel, particles enter with rate α, hop right with rate 1, exit with rate β.

Mean-field approximation

Assume no correlations between sites (eq 1 of the paper):

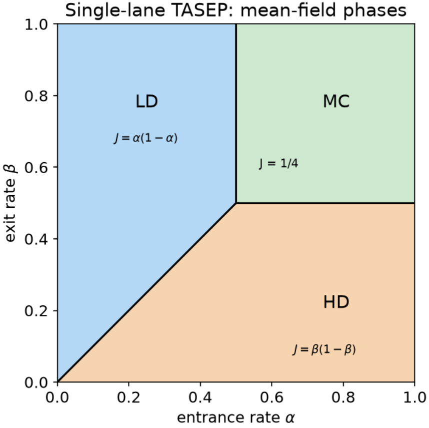
$$\langle p_i p_j \rangle = \langle p_i \rangle \langle p_j \rangle = p_i p_j .$$

The current across bond $i \to i{+}1$ is then

$$J = \langle p_i (1 - p_{i+1}) \rangle = \rho(1-\rho), \qquad \rho = \langle p_i \rangle .$$

Three stationary phases depending on which boundary is rate-limiting.

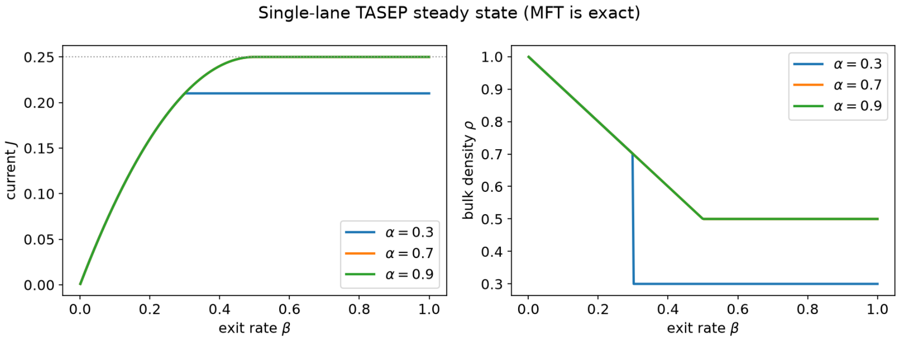
Single-lane phases and phase diagram



Phase	condition	J, ρ
LD	$\alpha < \beta$, $\alpha < 1/2$	$J{=}\alpha(1{-}\alpha)$, $\rho{=}\alpha$
HD	$\beta < \alpha$, $\beta < 1/2$	$J{=}\beta(1{-}\beta)$, $\rho{=}1{-}\beta$
MC	$\alpha > 1/2$, $\beta > 1/2$	$J{=}1/4$, $\rho{=}1/2$

For a single lane, **mean-field is exact** (matrix ansatz).

Single-lane currents



- At $\alpha > 1/2$, increasing β drives LD $\to$ MC through the maximal-current value $J = 1/4$.
- The current saturates: the bulk becomes the rate-limiting step.

Strategy: two single-lane TASEPs with effective rates

The two-channel model is two single-lane TASEPs (with exit rate β), coupled *only through the entrances* (eq 2 of the paper):

Effective entrance rates

$$\alpha_1 = \alpha \langle 1 - m_1 \rangle = \alpha(1 - m_1), \quad (1)$$

$$\alpha_2 = \alpha \langle 1 - p_L \rangle = \alpha(1 - p_L), \quad (2)$$

where m_1 = density at site 1 of channel 2 (its exit), p_L = density at site L of channel 1 (its exit).

Each channel is then a standard TASEP with rates (α_1, β) and (α_2, β) — whose steady states we already know.

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Nine candidate phases

3×3 combinations (LD/HD/MC per channel) $\rightarrow$ filter by self-consistency of α_1, α_2.

Symmetric phases: set-up

Symmetric ansatz (eq 6-7)

$$J_1 = J_2, \quad \rho_1 = \rho_2, \quad \alpha_1 = \alpha_2, \qquad p_i = m_{L-i+1}.$$

- Both channels in the same single-lane phase.
- Symmetry + the exit current relation determine the effective rates *self-consistently*.
- Key relation (exit current of channel 2, eq 8/11):

$$J_2 = \beta\, m_1 .$$

Symmetric MC phase (eqs 8–10)

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Step 1: current fixes m_1 (eq 8). In MC, $J_1 = J_2 = 1/4$; the exit current of channel 2 gives

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Step 3: existence condition (eq 10). MC needs $\alpha_1 > 1/2$ and $\beta > 1/2$:

$$\alpha\left(1 - \frac{1}{4\beta}\right) > \frac{1}{2} \quad \Longrightarrow \quad \alpha > \frac{2\beta}{4\beta - 1}$$

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$$\alpha_1 = \alpha(1-m_1) = \alpha\left(1 - \frac{\alpha_1(1-\alpha_1)}{\beta}\right).$$

A quadratic in α_1; physical root (eq 12):

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Conclusion

Symmetric HD phase is impossible in the two-channel model with narrow entrances. Only MC and LD survive as symmetric phases.

Asymmetric phases: MC/HD and MC/LD don't exist

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Conclusion

An asymmetric phase must pair **one channel in LD** with the other in a different phase: the candidate is HD/LD.

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Conditions (eq 20). Channel 1 in HD: $\alpha_1 > \beta$, $\beta < 1/2$; Channel 2 in LD: $\alpha_2 < \beta$, $\alpha_2 < 1/2$.

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$$J_1 = \beta p_L = \beta(1-\beta) \quad \Rightarrow \quad p_L = 1-\beta, \qquad J_2 = \beta m_1 = \alpha_2(1-\alpha_2).$$

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Step 2: effective rates (eq 22).

$$\alpha_2 = \alpha(1-p_L) = \alpha\beta, \qquad \alpha_1 = \alpha(1-m_1) = \alpha(1-\alpha+\alpha^2\beta).$$

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Step 3: existence (eq 23). HD condition $\alpha_1 > \beta$:

$$\alpha(1-\alpha+\alpha^2\beta) > \beta \quad \Longrightarrow \quad \beta < \frac{\alpha}{1+\alpha+\alpha^2}$$

Asymmetric LD/LD phase: set-up (eqs 24–26)

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Currents and effective rates (eqs 25–26)

$$
\begin{aligned}
J_1 &= \beta p_L = \alpha_1(1-\alpha_1), & J_2 &= \beta m_1 = \alpha_2(1-\alpha_2), \\
p_L &= \frac{\alpha_1(1-\alpha_1)}{\beta}, & m_1 &= \frac{\alpha_2(1-\alpha_2)}{\beta}, \\
\alpha_1 &= \alpha\left(1-\frac{\alpha_2(1-\alpha_2)}{\beta}\right), & \alpha_2 &= \alpha\left(1-\frac{\alpha_1(1-\alpha_1)}{\beta}\right).
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$$\alpha_1 = \alpha\left(1 - \frac{\alpha_2(1-\alpha_2)}{\beta}\right), \qquad \alpha_2 = \alpha\left(1 - \frac{\alpha_1(1-\alpha_1)}{\beta}\right).$$

Two coupled quadratics in α_1, α_2. Simplify with symmetric and antisymmetric combinations (eq 27):

$$S = \alpha_1 + \alpha_2, \qquad D = \alpha_1 - \alpha_2 \neq 0.$$

Asymmetric LD/LD phase: solve (eqs 27–31)

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Since $D \neq 0$ (asymmetric phase), we get

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$$S = 2\alpha - \frac{\alpha}{\beta}\left[S - \frac{S^2 + D^2}{2}\right] \qquad \text{(using } \alpha_1^2 + \alpha_2^2 = (S^2 + D^2)/2\text{).}$$

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Step 3: substitute $S = 1 - \beta/\alpha$ (eq 31).

$$D^2 = 1 - 4\beta + 2\frac{\beta}{\alpha} - 3\left(\frac{\beta}{\alpha}\right)^2$$

Asymmetric LD/LD phase: effective rates (eq 32)

From S and D we reconstruct the two effective entrance rates:

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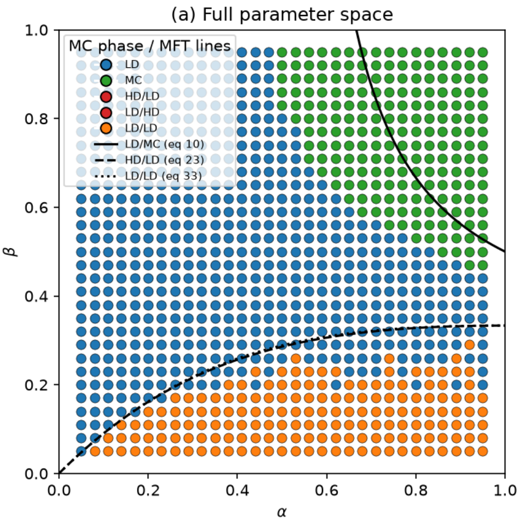
Existence (eq 33)

Real roots need $D^2 > 0$; together with the LD conditions (eq 24):

$$\frac{\alpha}{1+\alpha+\alpha^2} < \beta < \frac{\alpha(1-2\alpha) + 2\alpha\sqrt{\alpha^2 - \alpha + 1}}{3}$$

Mean-field phase diagram (summary)



Phase	MFT boundary
MC	$\alpha > \dfrac{2\beta}{4\beta - 1}$ (eq 10)
LD	$\alpha < \dfrac{2\beta}{4\beta - 1}$ (eq 13)
HD	never (needs $\alpha > 1$)
HD/LD	$\beta < \dfrac{\alpha}{1 + \alpha + \alpha^2}$ (eq 23)
LD/LD	eq 33 (narrow band)

Transitions: LD↔MC and LD↔LD/LD are *continuous*; HD/LD↔LD/LD is *first order*.

Take stock: what mean-field predicts

1. Two symmetric phases (LD, MC) — HD impossible.
2. Two symmetry-broken phases: HD/LD (first order) and LD/LD (continuous), in a *very narrow* band of β.
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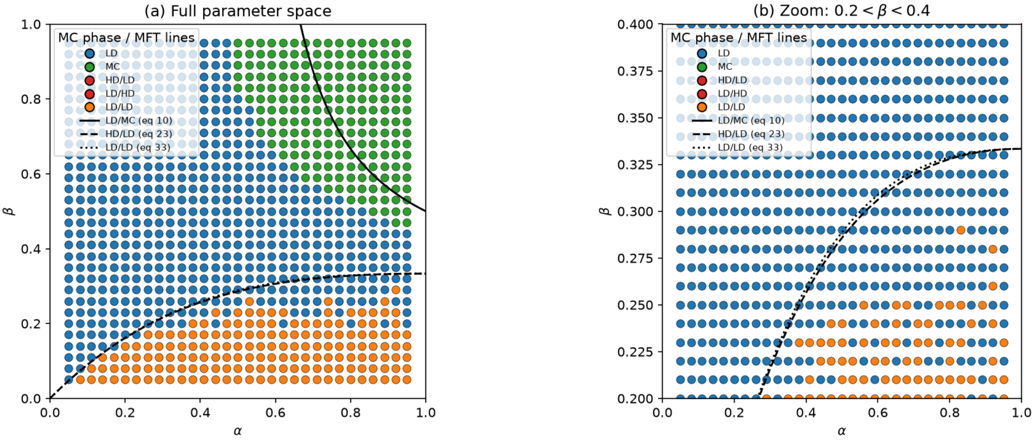
The message to verify

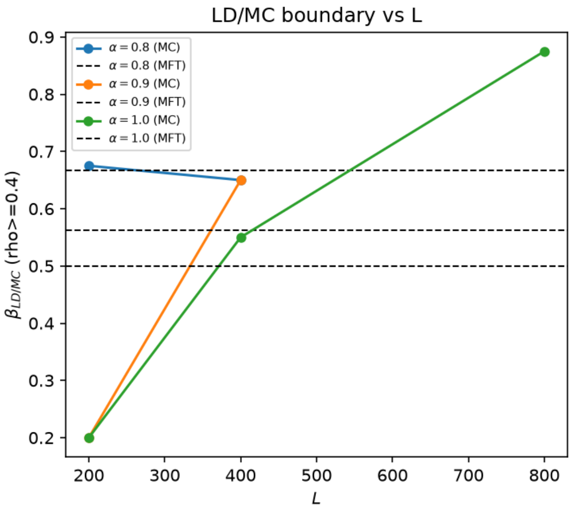
Mean-field predicts *where* transitions happen. Are these positions exact? For single-lane TASEP they are. Here, the boundary coupling may introduce correlations mean-field misses.

Phase diagram: MC vs MFT



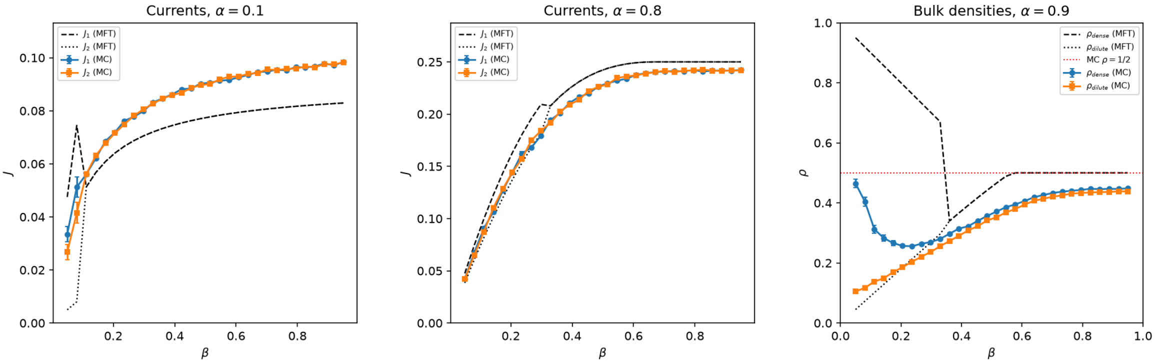
- Lines: mean-field; symbols: MC at $L = 1000$ (reproduced).
- Overall topology agrees; the quantitative position of boundaries **deviates**.

Transition positions: the paper's key message



- MFT predicts the LD/MC transition at $\beta_{LD/MC} = \frac{\alpha}{4\alpha-2}$ (eq 10).
- MC locates it at a **different** β — the deviation grows with α ($+0.01$ at $\alpha = 0.8 \to +0.075$ at $\alpha = 1.0$).
- Unlike single-lane TASEP, where MFT gives *exact* phase boundaries, here the transition positions are *approximate*.

Currents and densities



$\alpha = 0.1$ $\alpha = 0.8$ $\alpha = 0.9$

- Lines: mean-field; symbols: MC.
- Good agreement at low α, β; deviations grow with α, β.

Why does mean-field fail here?

The source of the discrepancy

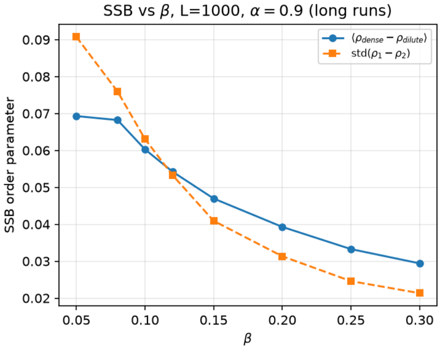
Mean-field assumes *no correlations* between sites (eq 1). The narrow-entrance coupling ties each entrance to the other channel's exit site — a strongly-correlated, local "impurity" at the boundary that mean-field cannot capture.

- These boundary correlations *propagate into the bulk* and shift the effective phase boundaries.
- The paper's own conclusion (Sec. 4): symmetry breaking in this model is a *boundary-induced* phenomenon — an effective impurity at the boundary.
- Whether the MFT-vs-MC deviation persists in the thermodynamic limit is still open (needs longer runs at $L = 800$).

SSB: the robust order parameter

Why a robust order parameter is needed

Time-averaged $|\rho_1 - \rho_2|$ is *washed out* because the broken states flip between $(+,-)$ and $(-,+)$ on long time scales. Robust choice: $\mathrm{std}(\rho_1 - \rho_2)$ over an ensemble of replicas.



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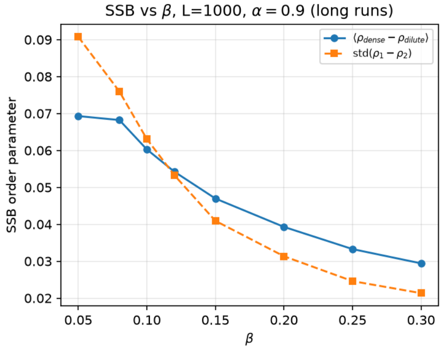
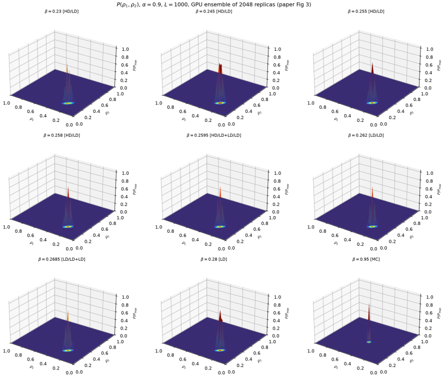


- Long-run scan at $L = 1000$, $\alpha = 0.9$: order parameter decreases monotonically with β ($0.09 \to 0.02$ from β=0.05 to 0.30).
- SSB confined to *low* β in our MC (vs the paper's $\beta \approx 0.23$ at the same L).

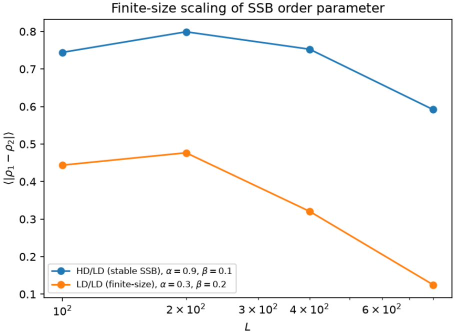
$P(\rho_1, \rho_2)$ from an ensemble (Fig 3)

A single short run at $L = 1000$ stays in *one* broken state (one peak). The paper's *two* peaks are recovered by an ensemble of independent replicas:

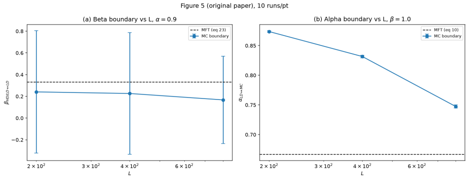


- Bimodal $P(\rho_1, \rho_2)$: two asymmetric HD/LD configurations.
- Animations sweeping β: 3D (mp4) and 2D (mp4).

Finite-size effects



SSB order parameter vs L



Phase boundaries vs L (original paper Fig 5)

- At $L = 200$: strong, immediate HD/LD (dense ~ 0.9, dilute ~ 0.04).
- At $L = 1000$: broken states flip; long runs are needed to observe the separation.
- Paper: the LD/LD band *shrinks* with L — likely a finite-size effect that vanishes in the thermodynamic limit.

Simulations: BKL and the GPU ensemble

BKL (Bortz–Kalos–Lebowitz)

- Rejection-free: pick events proportional to their rates.
- Classic double-loop selection: cost $\propto n_{\mathrm{active}}$.
- Fenwick-tree selection in $O(\log L)$: $\sim\mathbf{4}\times$ **faster, identical physics**.

Kernel	Msteps/s
classic BKL	1.2
Fenwick BKL	4.8

GPU: one thread per replica

$$\text{GPU}: \ \mathrm{rep}_1 \oplus \mathrm{rep}_2 \oplus \cdots \oplus \mathrm{rep}_N$$

- A serial chain maps poorly to the GPU; an *ensemble* maps perfectly.
- Thousands of replicas per β — exactly what $P(\rho_1, \rho_2)$ needs.

Backend	Msteps/s
CPU, 1 thread	4.4
CPU, 12 cores	29.6
GPU ensemble (2060 SUPER)	$\sim$85

One run $\rightarrow$ all figures

- `run_all_gpu.py`: one GPU ensemble scan produces ALL MC data (phase diagram, currents, densities, $P(\rho_1, \rho_2)$, SSB) saved to `results/gpu/*.npz`.
- `plot_all.py`: plots every figure by *reading* the saved data — no MC at plot time.
- Reproducible: fixed seeds, checkpointed chunks (crash-safe).

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Cost model (BKL vs Fenwick vs GPU)

Classic BKL $\propto n_{\text{active}}$ (fast at low density), Fenwick/GPU $\propto \log L$ (wins at high density). The paths *cross* in β.

Summary

- Reproduced the two-channel ASEP with narrow entrances: model, MFT phase diagram, currents, densities, $P(\rho_1, \rho_2)$.
- Mean-field predicts two symmetric (LD, MC) and two broken (HD/LD, LD/LD) phases via self-consistent effective entrance rates.
- Confirmed the paper's message: MFT transition positions are *not exact* — the boundary coupling creates correlations MFT misses.
- SSB characterized robustly with $\mathrm{std}(\rho_1 - \rho_2)$ and the ensemble $P(\rho_1, \rho_2)$; strong finite-size dependence.
- GPU ensemble ($\sim$85 Msteps/s) enables thousands of replicas and long runs.

Open questions

- Does the LD/LD band survive at $L = 1000$ and move with L? (Paper suggests it vanishes in the thermodynamic limit.)
- Does the MFT-vs-MC deviation shrink with L or persist (genuine mean-field failure)? Needs longer runs at $L = 800$.
- Can going beyond 2nd order / adding boundary-correlation parameters fix MFT?
- Extensions: asymmetric rates, wider entrances, 3 channels?

Thank you!